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JAVA PROJECT REPORT ON

PROJECT TITLE

Vehicle Parking Fee Collection and Report Generation System

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DECLARATION

I hereby declare that the project entitled "**Vehicle Parking Fee Collection and Report Generation System"** submitted in partial fulfillment of the requirements for the course [**Java Programming]** is a **record of my original work** carried out under the guidance of **[Mugilan]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Varshini.M

(Your name)

CERTIFICATE

This is to certify that the project entitled **"Vehicle Parking Fee Collection and Report Generation System "** is a **bonafide work** carried out by **[Varshini M] [Bsc.IT -B/2025-2028]**, [**Information Technology]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[Java Programming]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
(Technical Trainer, KGCAS)

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Head of the Department
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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The Vehicle Parking Fee Collection and Report Generation System is a software-based solution designed to automate and efficiently manage parking operations in areas such as shopping malls, offices, hospitals, and public parking spaces. Traditional parking systems often rely on manual processes, which can lead to errors in fee calculation, data mismanagement, and time inefficiency. This project aims to overcome these challenges by providing a computerized system that ensures accurate fee collection, real-time tracking, and detailed report generation.

The system is developed using Java programming language, with a user-friendly interface that allows operators to manage vehicle entry and exit smoothly. When a vehicle enters the parking area, the system records essential details such as vehicle number, type (two-wheeler, four-wheeler, etc.), entry time, and generates a unique parking ID. Upon exit, the system calculates the parking fee automatically based on the duration of stay and predefined pricing rules.

One of the key features of this system is its ability to generate various reports. These reports include daily, weekly, and monthly revenue summaries, vehicle entry/exit logs, and occupancy statistics. Such reports help administrators analyze parking usage, monitor income, and make better management decisions. The system also reduces the chances of human errors and improves transparency in fee collection.

Additionally, the system ensures data security and easy retrieval of records. It can store large amounts of parking data and allows quick searching of vehicle details when required. The implementation of this system not only enhances operational efficiency but also provides a scalable and reliable solution for modern parking management needs.

1. INTRODUCTION

In the modern world, the number of vehicles has increased rapidly due to urbanization and economic growth. As a result, managing parking spaces has become a major challenge in cities, shopping malls, hospitals, educational institutions, and office complexes. Traditional parking systems are mostly manual, where attendants record vehicle details on paper and calculate parking fees manually. This method is time-consuming, error-prone, and inefficient, especially during peak hours.

To overcome these issues, the Vehicle Parking Fee Collection and Report Generation System has been developed as an automated solution. This system is designed to efficiently manage parking operations by using computer-based technology. It simplifies the process of vehicle entry and exit, calculates parking fees automatically, and maintains accurate records of all transactions.

The system is implemented using the Java programming language, which provides a reliable and platform-independent environment for developing such applications. When a vehicle enters the parking area, the system records important details such vehicle number, type of vehicle, and entry time. Each vehicle is assigned a unique identifier to track its parking activity. When the vehicle exits, the system calculates the parking fee based on the duration of stay and predefined rates. This ensures accurate billing and eliminates manual calculation errors.

Another important feature of the system is report generation. The system can generate various reports such as daily, weekly, and monthly parking summaries. These reports include information about the number of vehicles parked, total revenue collected, and parking usage patterns. Such reports are very useful for administrators to monitor performance and make better decisions regarding parking management.

1.1 Objective

The main objective of the Vehicle Parking Fee Collection and Report Generation System is to develop an efficient and automated system for managing parking operations. This system aims to replace the traditional manual parking method with a computerized solution that ensures accuracy, speed, and reliability in handling vehicle parking activities.

The specific objectives of this project are:

-> To develop a computerized system for managing vehicle parking efficiently.

-> To record vehicle entry details such as vehicle number, type, and entry time accurately.

-> To track vehicle exit details and calculate the total parking duration.

-> To automatically calculate parking fees based on predefined rates and time duration.

-> To reduce manual errors in fee calculation and record maintenance

1.2 Overview

The Vehicle Parking Fee Collection and Report Generation System is a software application designed to manage and automate parking operations in an efficient and organized manner. The system replaces traditional manual methods with a computerized approach, ensuring accuracy, speed, and reliability in handling vehicle parking activities.

This system works by recording all necessary details when a vehicle enters the parking area. These details include the vehicle number, type of vehicle (such as two-wheeler or four-wheeler), and the exact entry time. Each vehicle is tracked throughout its parking duration using these details. When the vehicle exits, the system calculates the total time spent in the parking area and automatically determines the parking fee based on predefined pricing rules.

The system is developed using the Java programming language, making it platform-independent and easy to maintain. It uses simple logic and data structures to store vehicle information and perform operations such as entry, exit, fee calculation, and report generation. The application can be implemented as a console-based or graphical user interface (GUI) system depending on requirements.

One of the important features of this system is its ability to maintain a proper database of all parking transactions. It stores records of vehicles, entry and exit times, and collected fees. These records can be retrieved easily whenever needed, ensuring better data management and transparency.

1.3 Features

The **Vehicle Parking Fee Collection and Report Generation System** includes the following main features:

-> **Vehicle Entry Management**

Records vehicle details such as vehicle number, type, and entry time.

-> **Vehicle Exit Management**

Tracks vehicle exit and calculates total parking duration.

-> **Automatic Fee Calculation**

Calculates parking charges based on time and predefined rates.

-> **Unique Vehicle Identification**

Assigns unique ID or uses vehicle number for tracking.

-> **Data Storage and Managemen**t

Stores all parking records for future reference.

2. PROGRAM ANALYSIS

Program Analysis is the process of understanding the existing system, identifying its problems, and defining the requirements for the new system. In this project, the analysis focuses on improving the traditional parking system by introducing automation. It helps in identifying the limitations of manual parking management and designing a more efficient computerized solution.

2.1 Problem Definition

The problem addressed in this project is the inefficiency of traditional manual parking management systems. In many parking areas such as malls, offices, and hospitals, vehicle entry and exit details are recorded manually, and parking fees are calculated by hand.

Some common problems include:

-> Manual entry of vehicle details is time-consuming

-> High chances of human errors in fee calculation

-> Difficulty in maintaining and storing records

-> Loss or damage of data due to paper-based system

-> No proper way to track vehicle entry and exit

The Vehicle Parking Fee Collection and Report Generation System is designed to overcome these problems by providing an automated and reliable solution.

2.2 Project Overview

This project is a Java-based application that automates parking operations. It manages vehicle entry and exit, calculates parking fees, and stores all necessary data in an organized manner.

The Vehicle Parking Fee Collection and Report Generation System is a software application developed to automate and manage parking operations efficiently. The system is designed using the Java programming language and aims to replace manual parking methods with a computerized solution.

The system includes the following functionalities:

-> When a vehicle enters, its details are recorded

-> Entry time is stored automatically

-> When the vehicle exits, the system calculates the duration

-> Fee is calculated based on the duration

-> Data is stored for future reference

-> Reports are generated for analysis

2.3 Scopes and Limitations

Scope

The scope of the **Vehicle Parking Fee Collection and Report Generation System** includes the following:

-> When a vehicle enters, its details are recorded

-> Entry time is stored automatically

-> When the vehicle exits, the system calculates the duration

-> Fee is calculated based on the duration

-> Data is stored for future reference

-> Reports are generated for analysis

The system can be used in:

-> **Shopping Malls and Complexes**

To manage large numbers of vehicles and reduce congestion during peak hours.

-> **Office Buildings and IT Parks**

To organize employee and visitor parking efficiently.

-> **Hospitals and Clinics**

To handle continuous vehicle flow and ensure smooth parking operations.

-> **Airports and Railway Stations**

To manage high-volume parking with accurate fee collection.

3. SYSTEM DESIGN

System Design is an important phase in software development where the overall structure and working of the system are planned. It defines how the system will operate, how different components interact, and how data flows within the system. The Vehicle Parking Fee Collection and Report Generation System is designed to provide an efficient, reliable, and user-friendly solution for managing parking operations.

3.1 Modules

The main modules of the **Vehicle Parking Fee Collection and Report Generation System** are explained below:

1. Vehicle Entry Module

This module is responsible for collecting vehicle details when a vehicle enters the parking area.

Functions:

-> Accept vehicle number from the user

-> Accept vehicle type (Bike / Car / Other)

-> Record entry time or parking hours

Working:

-> When the user selects the “Add Vehicle” option, the system prompts for:

-> Vehicle number

-> Vehicle type

-> Number of hours parked

-> The entered details are stored temporarily in the system using objects or arrays.

2.Fee Calculation Module

This module calculates the parking fee based on vehicle type and parking duration.

Functions:

-> Determine rate per hour

-> Multiply hours with rate

Generate total fee

Example Rates:

-> Bike → ₹10 per hour

-> Car → ₹20 per hour

3. Data Storage Module

This module stores all parking details temporarily during program execution.

Functions:

-> Store vehicle number

-> Store vehicle type

-> Store hours and fee

Working:

-> Uses ArrayList (Java Collection) to store multiple vehicle records

-> Each vehicle is stored as an object

4.Time Management Module

This module manages entry and exit time of vehicles.

Functions:

-> Store entry time

-> Calculate total duration

-> Convert time into hours

5.Vehicle Exit Module

This module handles the process when a vehicle leaves the parking area.

Functions:

-> Record exit time

-> Calculate total parking duration

-> Display final fee

Working:

-> When a vehicle exits, the system:

-> Retrieves its entry details

-> Calculates total time stayed

-> Displays the final bill

4. METHODOLOGY

Methodology describes the step-by-step process followed to develop the system. It explains how the problem is analyzed, how the system is designed, implemented, tested, and finally documented.

For this project, a structured approach was followed to ensure the system works efficiently and meets all requirements.

The development process includes the following stages:

-> Problem Analysis

-> System Design

-> Program Development

-> Testing and Validation

-> Result and Documentation

-> Algorithm

-> Flowchart

4.1 Problem Analysis

In this phase, the existing manual parking system was studied carefully to identify its drawbacks.

Problems Identified:

-> Manual fee calculation leads to errors

-> Time-consuming process

-> Difficulty in managing multiple vehicles

-> No proper record maintenance

4.2. System Design

In this stage, the overall structure of the system is designed before coding begins.

The system is divided into modules such as:

-> Vehicle Entry Module

-> Fee Calculation Module

-> Data Storage Module

-> Report Generation Module

-> Exit Module

4.3. Program Development

After designing the system, the program is implemented using the **Java programming language**.

The following steps were followed during development:

-> Create a Vehicle class

-> Implement methods for fee calculation

-> Write menu-driven logic

-> Store data using ArrayList

-> Display output in proper format

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Enter vehicle details	Data stored correctly
Enter hours	Correct fee calculated
view report	All vehicles displayed
Invalid input	Error handled properly

4.5. Result and Documentation

The Result and Documentation phase is the final stage of the project development process. In this phase, the output of the system is verified, analyzed, and properly documented. It ensures that the system meets all the requirements and functions correctly without errors.

Proper documentation is prepared to explain the design, implementation, and working of the system.

4.6. Algorithm

Algorithm for the **Vehicle Parking Fee Collection and Report Generation System:**

1.Start the program

2.Display menu

3.Enter vehicle details

4.Select vehicle type

5.Enter parking hours

6.Calculate parking fee

7.Store vehicle data

8.Display report if required

9.Exit the program

10.End

4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

Start

↓

Display Menu

↓

Enter Vehicle Details

↓

Calculate Fee

↓

Store Data

↓

Display Report

↓

End

Students can draw using:

- draw.io
- Lucidchart
- MS Visio

5. SOURCE CODE

```
import java.util.Scanner;
public class ParkingSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String number = "", type = "";
        int hours = 0;
        double fee = 0;
        int choice;
        do {
            System.out.println("\n===== Parking System =====");
            System.out.println("1. Add Vehicle");
            System.out.println("2. View Report");
            System.out.println("3. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
 // ADD VEHICLE
            if (choice == 1) {
                System.out.print("Enter Vehicle Number: ");
                number = sc.next();
                System.out.print("Enter Type (bike/car/bus): ");
                type = sc.next
                System.out.print("Enter Hours: ");
                hours = sc.nextI
                if (type.equalsIgnoreCase("bike")) {
                    fee = 10 * hours;
```

```
            } else if (type.equalsIgnoreCase("car")) {
                fee = 20 * hours;
            } else if (type.equalsIgnoreCase("bus")) {
                fee = 30 * hours;
            } else {
                fee = 15 * hours;
            }
            System.out.println("Vehicle Added!");
        }
// VIEW REPORT
        else if (choice == 2) {
            if (number.equals("")) {
                System.out.println("No data available!");
            } else {
                System.out.println("\n--- Report ---");
                System.out.println("Vehicle Number: " + number);
                System.out.println("Type: " + type);
                System.out.println("Hours: " + hours);
                System.out.println("Fee: ₹" + fee);
            }
        }
    } while (choice != 3);
        System.out.println("Thank you!");
     sc.close();
  }
}
```

6. INPUT AND OUTPUT SCREENS

```
Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KGCAS\Desktop\java project>javac ParkingSystem.java

C:\Users\KGCAS\Desktop\java project>java ParkingSystem
```

```
===== Parking System =====
1. Add Vehicle
2. View Report
3. Exit
Enter choice: 1
Enter Vehicle Number: 3438
Enter Type (bike/car/bus): bike
Enter Hours: 3
Vehicle Added!

===== Parking System =====
1. Add Vehicle
2. View Report
3. Exit
Enter choice: 2

--- Report ---
Vehicle Number: 3438
Type: bike
Hours: 3
Fee: ?30.0

===== Parking System =====
1. Add Vehicle
2. View Report
3. Exit
Enter choice: 3
Thank you!
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The Vehicle Parking Fee Collection and Report Generation System was developed to overcome the limitations of traditional manual parking systems. During the development and implementation of this project, it was observed that automation significantly improves the efficiency and accuracy of parking management.

The system successfully records vehicle entry and exit details, calculates parking fees automatically, and stores all data in an organized manner. This reduces the chances of human errors that commonly occur in manual systems. It also minimizes the time required for processing vehicles, especially during peak hours, thereby improving user convenience.

The report generation feature plays an important role in analyzing parking activities. It provides useful insights such as total vehicles parked, total revenue collected, and usage patterns. These reports help administrators in monitoring system performance and making informed decisions.

7.2 Conclusion

In conclusion, the Vehicle Parking Fee Collection and Report Generation System provides an efficient and automated solution for managing parking activities. It replaces the traditional manual system with a computerized approach that ensures accuracy, speed, and reliability.

The system simplifies the process of vehicle entry and exit, reduces manual work, and ensures proper record maintenance. Automatic fee calculation eliminates errors and ensures fair billing, while the report generation feature helps in better analysis and decision-making.

This project demonstrates the practical application of software technology in solving real-world problems. It improves operational efficiency, saves time, and enhances user satisfaction. With further improvements, the system can be expanded to handle larger parking areas and advanced functionalities.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com